



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO P. M. CHINNASAMI FUEL STATION

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

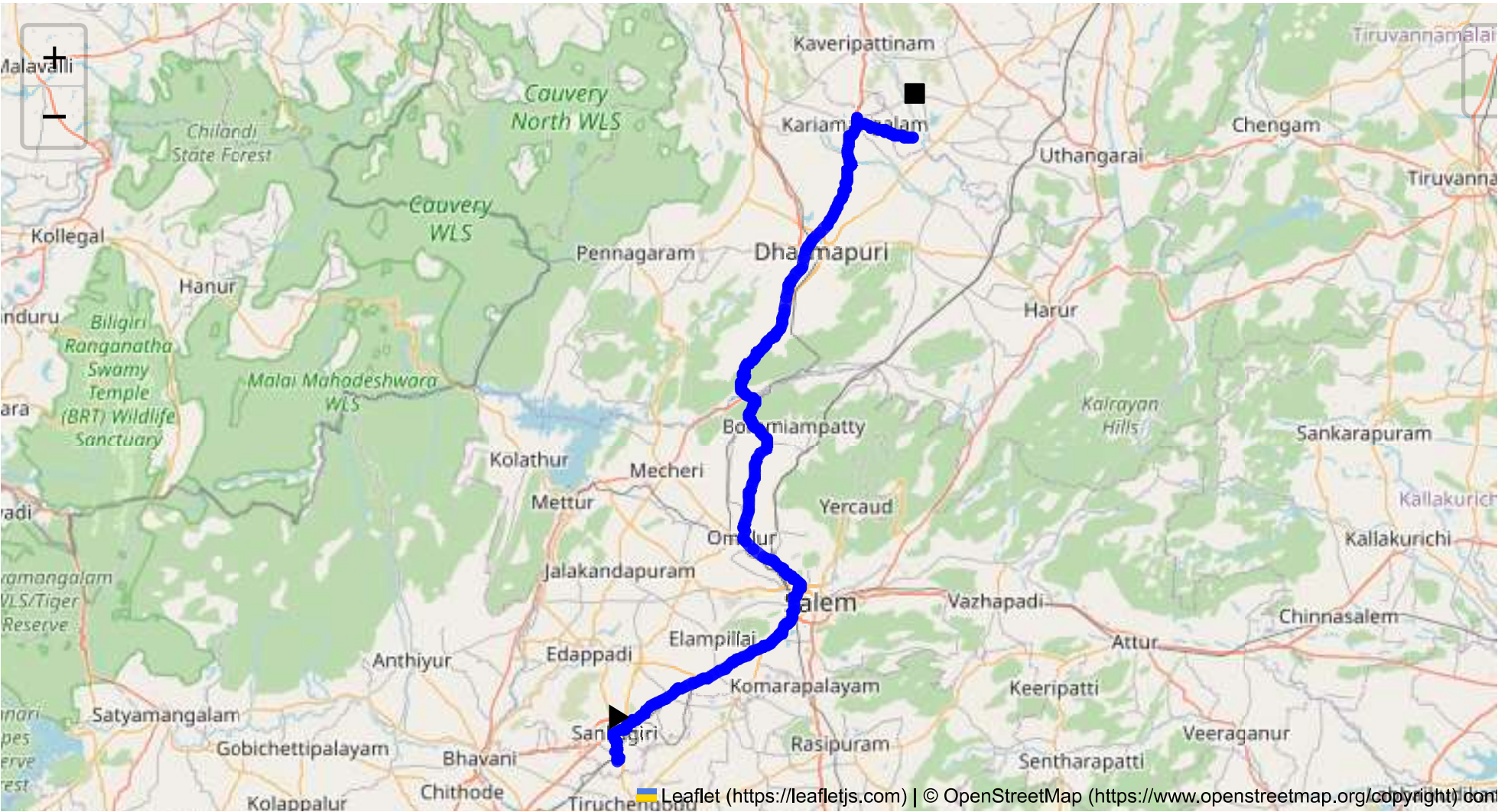
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 137.22 km**  
**Estimated Duration: 2.6 hours**  
**Adjusted Duration (Heavy Vehicle): 3.2 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.288282, 78.286353)**



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Chellakuttapatti covers approximately 137.22 kilometers. It primarily includes state highways and potentially national highways, with a travel time of around 2.59 hours for heavy vehicles. The journey takes you through rural and semi-urban areas, with varying road conditions.



# Weather Conditions

Tamil Nadu typically experiences a tropical climate. The region can face heavy rains during the monsoon season (June to September) and extremely high temperatures in the summer months (March to May). Potential weather-related hazards include:

- **Monsoon:** Flooded roads, reduced visibility, possible landslides in hilly areas.
- **Summer:** Tire blowouts due to high temperatures, potential dehydration for drivers.

# Traffic Patterns

Traffic density can vary, with peak hours generally in the morning (7 AM to 10 AM) and late afternoon/evening (5 PM to 8 PM). Major congestion areas may be around towns or intersections along the route. Regular updates from local traffic authorities can provide more specific congestion-prone areas.

# Road Quality and Infrastructure

- **Highways:** Generally good quality, but look out for occasional potholes or uneven surfaces.
- **Rural areas:** May have narrow roads, limited signage, and less maintenance.
- **Bridges:** Ensure to check weight limitations, especially for heavy vehicles.

# Alternative Routes

In case of emergencies, consider the following:

- **State highways:** Look for parallel state highways that might offer alternative passage.
- **Local roads:** These could be viable for detours but often require caution due to lower maintenance levels.

# Local Regulations

Tamil Nadu has specific regulations regarding hazardous material transport, such as route permit requirements and time restrictions to ensure safety. It is important to comply with the designated routes and permissible hours of transit. The driver should have all necessary documentation and permits.

# Historical Incidents

While data on specific historical incidents should be checked with local authorities, it is prudent to maintain vigilance in areas previously known for accidents, especially near sharp turns, intersections, or dense traffic zones.

# Environmental Considerations

- **Sensitive Areas:** Watch for regions marked as forest reserves or wildlife sanctuaries. Avoid off-road driving in these areas.
- **Pollution Control:** Ensure vehicle compliance with emission norms to protect local environments.

# Communication Coverage

Mobile network coverage in urban areas is generally reliable; however, rural or remote sections of the route might experience dead zones. Satellite communication or offline maps could serve as reliable backups.

# Emergency Response Times

- **Urban Areas:** Swift response typically within 30 minutes to an hour.
- **Rural Areas:** Delays possible, ranging from 1 to 2 hours based on the distance to the nearest town.
- **Accident-Prone Spots:** Be cautious in areas with a history of accidents, as response might be quicker here due to pre-stationed services.

# Overall Risk Assessment

The route poses moderate risks mainly due to weather conditions, variable road quality, and potential congestion at peak hours. Ensuring thorough preparation, adherence to regulations, and awareness of the local environment can mitigate many risks. Communication and responsiveness strategies should be in place to handle any emergencies efficiently.

## Risk Assessment - Turns

|    | Risk Type | Risk Level | Coordinates        | Speed Limit |
|----|-----------|------------|--------------------|-------------|
| 0  | Turn      | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1  | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2  | Turn      | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3  | Turn      | High       | 11.45028, 77.87202 | 15 KM/Hr    |
| 4  | Turn      | Medium     | 11.45309, 77.87325 | 30 KM/Hr    |
| 5  | Turn      | Medium     | 11.46400, 77.87323 | 30 KM/Hr    |
| 6  | Turn      | Medium     | 11.46765, 77.87037 | 30 KM/Hr    |
| 7  | Turn      | High       | 11.47403, 77.86841 | 15 KM/Hr    |
| 8  | Turn      | High       | 11.49149, 77.89601 | 15 KM/Hr    |
| 9  | Turn      | High       | 11.49224, 77.89601 | 15 KM/Hr    |
| 10 | Turn      | Medium     | 11.53501, 77.95707 | 30 KM/Hr    |
| 11 | Turn      | Medium     | 11.53601, 77.95856 | 30 KM/Hr    |
| 12 | Turn      | Medium     | 11.62403, 78.10893 | 30 KM/Hr    |
| 13 | Turn      | High       | 11.64180, 78.11808 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 14 | Turn       | High       | 11.64325, 78.11993 | 15 KM/Hr    |
| 15 | Turn       | Medium     | 11.69519, 78.10775 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.73958, 78.05105 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.75835, 78.05016 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.86847, 78.08367 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.87942, 78.08197 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.92585, 78.06507 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.93704, 78.05402 | 30 KM/Hr    |
| 22 | Turn       | High       | 11.95818, 78.04978 | 15 KM/Hr    |
| 23 | Turn       | Medium     | 11.95988, 78.04948 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.96304, 78.05136 | 30 KM/Hr    |
| 25 | Turn       | High       | 11.96399, 78.05135 | 15 KM/Hr    |
| 26 | Turn       | High       | 11.96580, 78.04923 | 15 KM/Hr    |
| 27 | Turn       | Medium     | 11.96675, 78.04939 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.96791, 78.05070 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 12.19881, 78.18157 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 12.22459, 78.19136 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 12.29110, 78.19909 | 30 KM/Hr    |
| 32 | Blind Spot | Blind Spot | 12.31460, 78.20761 | 10 KM/Hr    |
| 33 | Turn       | High       | 12.30795, 78.20891 | 15 KM/Hr    |
| 34 | Turn       | High       | 12.29976, 78.24628 | 15 KM/Hr    |
| 35 | Turn       | Medium     | 12.29890, 78.24676 | 30 KM/Hr    |
| 36 | Turn       | Medium     | 12.29768, 78.25031 | 30 KM/Hr    |

Emergency Locations

|   | type     | name               | coordinates               | speed_limit | risk_level |
|---|----------|--------------------|---------------------------|-------------|------------|
| 0 | hospital | Thilagam Hospital  | 11.4753573,<br>77.8685829 | 30 km/h     | Medium     |
| 1 | clinic   | Meyyazhagan Clinic | 11.4752254,<br>77.8685671 | 30 km/h     | Medium     |

|    | type     | name                                                            | coordinates               | speed_limit | risk_level |
|----|----------|-----------------------------------------------------------------|---------------------------|-------------|------------|
| 2  | hospital | Dr. Jaganathan Hospital                                         | 11.4752717,<br>77.8709312 | 30 km/h     | Medium     |
| 3  | hospital | Kathir Eye Hospital                                             | 11.4740976,<br>77.8681591 | 30 km/h     | Medium     |
| 4  | police   | Police Quarters                                                 | 11.4741243,<br>77.8673495 | 30 km/h     | Medium     |
| 5  | hospital | Sankari Hospital                                                | 11.4777496,<br>77.873892  | 30 km/h     | Medium     |
| 6  | clinic   | Dr. Dhanasekar Clinic                                           | 11.4793113,<br>77.8829247 | 30 km/h     | Medium     |
| 7  | hospital | Sri Vellaiyan Hospital                                          | 11.4798876,<br>77.8831394 | 30 km/h     | Medium     |
| 9  | hospital | C. N. Hospital                                                  | 11.55724, 78.0096         | 30 km/h     | Medium     |
| 10 | hospital | Vinayaka Mission's Homeopathic Medical College & Hospital       | 11.583617,<br>78.059145   | 30 km/h     | Medium     |
| 11 | hospital | Vinayaka mission Kirubananadah variyar medical college hospital | 11.584984,<br>78.0618417  | 30 km/h     | Medium     |
| 12 | hospital | vims                                                            | 11.5854239,<br>78.0639188 | 30 km/h     | Medium     |
| 15 | hospital | Vinayaka Mission’s Kirupananda Variyar Medical College          | 11.5977669,<br>78.0850992 | 30 km/h     | Medium     |
| 19 | hospital | Bharathi Hospital, Salem                                        | 11.6492647,<br>78.1204131 | 30 km/h     | Medium     |
| 20 | hospital | Sisu Hospital Inida Ltd                                         | 11.6505276,<br>78.1205003 | 30 km/h     | Medium     |
| 21 | hospital | Aishwarya Hospital                                              | 11.6716291,<br>78.1267807 | 30 km/h     | Medium     |
| 22 | hospital | C.M. Hospital                                                   | 11.6747345,<br>78.1255034 | 30 km/h     | Medium     |
| 24 | pharmacy | Healthy Pharmacy                                                | 11.6762714,<br>78.127744  | 30 km/h     | Medium     |
| 25 | pharmacy | Thulasi Pharmacy                                                | 11.6761991,<br>78.1277377 | 30 km/h     | Medium     |
| 26 | hospital | Sri Gokulam Speciality Hospital                                 | 11.6788724,<br>78.1274258 | 30 km/h     | Medium     |
| 27 | hospital | Global Medical Center                                           | 11.6867049,<br>78.1186697 | 30 km/h     | Medium     |
| 28 | hospital | Manipal Hospital                                                | 11.703072,<br>78.1026612  | 30 km/h     | Medium     |

|    | type     | name                               | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------|---------------------------|-------------|------------|
| 31 | clinic   | Manimuthu Skin Clinic              | 11.7521189,<br>78.0484058 | 30 km/h     | Medium     |
| 32 | hospital | St. Joseph's Speciality Hospital   | 11.761611,<br>78.0522344  | 30 km/h     | Medium     |
| 33 | hospital | Anugraha Multi Speciality Hospital | 11.773549,<br>78.0546408  | 30 km/h     | Medium     |
| 34 | hospital | L.R. Hospital                      | 11.8703354,<br>78.0844554 | 30 km/h     | Medium     |
| 35 | clinic   | JSA Clinic                         | 11.8700619,<br>78.0844982 | 30 km/h     | Medium     |
| 37 | hospital | Government Hospital                | 12.3048002,<br>78.2060881 | 30 km/h     | Medium     |

Crowded Spots

|    | type       | name                                                       | coordinates               | speed_limit | risk_level |
|----|------------|------------------------------------------------------------|---------------------------|-------------|------------|
| 13 | university | Vinayaka Missions Vinayaka Missions<br>Research Foundation | 11.5974056,<br>78.0843617 | 30 km/h     | Medium     |
| 14 | school     | Vinayaka Missions Annapoorani Matric<br>School             | 11.5971662,<br>78.0839876 | 30 km/h     | Medium     |
| 16 | school     | Dist Education Training                                    | 11.6133799,<br>78.1022409 | 30 km/h     | Medium     |
| 17 | school     | DIET - Salem                                               | 11.6144158,<br>78.1025078 | 30 km/h     | Medium     |
| 18 | school     | Diamond Rays Matric HR Sec School                          | 11.6385629,<br>78.1206259 | 30 km/h     | Medium     |
| 23 | college    | Sona College                                               | 11.6768,<br>78.1257052    | 30 km/h     | Medium     |
| 29 | university | Periyar University                                         | 11.7186758,<br>78.0771089 | 30 km/h     | Medium     |
| 30 | school     | Fathima Hr Sec School                                      | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 36 | college    | Laxminarayana Polytechnic College                          | 12.0421277,<br>78.103203  | 30 km/h     | Medium     |
| 38 | school     | Government Boys Higher secondary<br>school                 | 12.3047059,<br>78.2065495 | 30 km/h     | Medium     |



## Route Photos of Risky Spots



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44900, 77.87406





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45028, 77.87202



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45309, 77.87325





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46400, 77.87323



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46765, 77.87037



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.47403, 77.86841





Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.49149, 77.89601



Risk Type: Turn  
Risk Level: High  
Speed Limit: 15 KM/Hr  
Coordinates: 11.49224, 77.89601





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53501, 77.95707



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.53601, 77.95856



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.62403, 78.10893





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.64180, 78.11808



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.64325, 78.11993





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.69519, 78.10775



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.73958, 78.05105



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.75835, 78.05016





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86847, 78.08367



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.87942, 78.08197





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.92585, 78.06507**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.93704, 78.05402**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.95818, 78.04978**





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95988, 78.04948



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96304, 78.05136





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96399, 78.05135



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96580, 78.04923



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96675, 78.04939





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96791, 78.05070



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.19881, 78.18157





Coordinates: 12.22459, 78.19136



Coordinates: 12.29110, 78.19909



Coordinates: 12.31460, 78.20761





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.30795, 78.20891



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.29976, 78.24628





Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29890, 78.24676



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29768, 78.25031